Problem Solving and Calculations

Set 4: Gravitation and Satellites

Useful Data
Mass of Earth: 5.98×10^{24} kg
Mass of Moon: 7.34×10^{22} kg
Mass of Sun: 1.99×10^{30} kg

Radius of Earth: 6.37×10^{6} m
g at sea level: 9.80 m s^{-2}

Notes

1. If all objects are attracted to one another by gravity, why are you not attracted towards large buildings?

2. a) When mountaineers climb Mount Everest to a height of about 8 km, does their weight change? Explain your answer.
   b) How does the weight of an underground miner change as he descends into a very deep mine? Explain your answer.
   c) A geophysicist's assistant measures the acceleration due to gravity with a very sensitive accelerometer at various places on the Earth's surface and finds that it is slightly different at each place. What are two possible reasons for this variation?

3. What is the weight of a free falling object? Explain your answer.

4. If air resistance is negligible, all free falling objects near the Earth's surface accelerate at the same rate even though they have different masses. Explain.

5. Black holes form when massive stars at least four times the mass of the Sun collapse into a tiny fraction of their original volume. Why is the gravitational force near a black hole so large that not even light can escape from it, even though its mass is the same as that of the original star?

6. A student measures the acceleration due to gravity at the Earth's surface and finds that it is 10 m s^{-2}. Calculate the mass of the Earth according to the student's result.

7. In an experiment to measure the force of gravity, a physicist suspends two 100 kg lead spheres so that their centres are 622 mm apart. Calculate the force of attraction between the spheres.

8. The gravitational force acting on the space shuttle at sea level is F.
   a) At what height above the Earth's surface would the gravitational force acting on the shuttle be ½F?
   b) The shuttle is in orbit around the Earth at a height of 610 km above the Earth's surface. What is the gravitational acceleration the shuttle experiences?
   c) The Space Shuttle Discovery carried the Hubble Space Telescope to an altitude of 610 km. What orbital speed did the shuttle have to give the telescope to keep it in this orbit?

9. The Earth attracts the Moon with a force of 2.03×10^{20} N. Use this information to calculate how far the Moon is from the Earth.

10. Neptune has a mass 16.6 times that of the Earth and its radius is 3.89 times the Earth's radius. Compare the gravitational field strength at Neptune's surface with that at the Earth's surface.

11. a) The Moon is in an almost circular orbit around the Earth. Why doesn't the gravitational force of the Earth acting on the Moon change the Moon's speed?
    b) Calculate the net force on the Moon due to the Earth and the Sun during a solar eclipse. The radius of the Moon's orbit is 3.80×10^{8} m and the radius of the Earth's orbit is 1.49×10^{11} m.

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Notes

12. Which is greater, the period of revolution of a communications satellite at a height in excess of 30 000 km, or that of a satellite that orbits at a height of a few hundred kilometres? Show your reasoning.

13. a) A research satellite in low Earth orbit move across the Earth's surface at several kilometres per second. If engineers fired retro-rockets on board this satellite and slowed it down very quickly, what would happen to the satellite's orbit?
    b) Communications satellites are always located above the same spot on the Earth's surface, that is, their speed across the Earth's surface is zero. Why do they remain in orbit?

14. a) Why do engineers launch rockets carrying satellites in an easterly direction?
    b) Why do engineers try to locate launch facilities as close to the equator as they can?

15. Scientists aboard an orbiting space station want to return some equipment to Earth in a capsule. What must they do with the capsule to achieve this?

16. Technicians want to move a satellite in a stable circular orbit to a lower orbit. Engineers achieve this by reducing the satellite's speed and therefore reducing its kinetic energy. As it moves to a lower orbit why does its speed increase?

17. Astronauts in orbit in a space station float around unless they are strapped into their seats. Why?

18. Calculate the period of a satellite orbiting the Earth at a height of 550 km.

19. The moon Europa orbits Jupiter at a radius of 6.71×10^8 m and an orbital period of 3.07×10^5 s.
    a) Calculate the Europa's orbital speed.
    b) Calculate Jupiter's mass.

20. Scientists have used data from the motions of satellites to determine accurately Earth's mass. A navigation satellite in a circular orbit 2.02×10^4 km above the surface has a period of 12.0 h. From this information, calculate the Earth's mass.

21. What is the height above the Earth's surface of a communications satellite if it always orbits above a particular spot on the equator?

22. Titan is one of Saturn's moons. It completes one orbit every 14.0 Earth days. Our Moon circles the Earth in 27.3 Earth days. The mass of Saturn is equal to 108 Earth masses. Find the ratio of Titan's orbital radius to that of our Moon.

23. The orbit of Mercury is elliptical. Its perihelion (closest approach to the Sun) is 4.60×10^{10} m and its aphelion, or furthest distance from the Sun, is 6.90×10^{10} m.
    a) Calculate the force of attraction between the Sun and Mercury at each position.
    b) What other parameter of Mercury's orbit changes? Calculate this quantity for the maximum and minimum distances from the Sun.
    c) Explain how this can happen without Mercury losing energy and crashing into the Sun.

Set 3	5	$0.82\ m\ s^{-2}$ toward the centre of the circle
	6	61.9 N
	7 a)	$1.54\ m\ s^{-2}$
	7 b)	17.5 N
	8	17.6°
	9 a)	691 N
	9 b)	3.55 s
	10 a)	yes
	10 b)	3.09 kN
	10 c)	15.3°
	11	$15.8\ m\ s^{-1}$
	12 a)	$37.8\ m\ s^{-1}$
	12 b)	$1.90 \times 10^{4}\ m\ s^{-2}$
	12 c)	17.7 kHz
	13 b)	2.59×10^{-16} N
	13 d)	5.53×10^{8} m
	13 e)	it would drop 30.6 m
	16 a)	$88.5\ m\ s^{-1}$
	b)	$198\ m\ s^{-1}$
	17	$87.4\ m\ s^{-1}$
	18 a)	180 N upward (b) 20.4 N downward
	19 a)	20 m
	20 b)	$4.32\ m\ s^{-1}$
	c)	2.00×10^{3} N
	21 b)	$5.94\ m\ s^{-1}$
	c)	at top, 0; at bottom, 1176 N upward
	d)	at top, 441 N upward; at bottom, 735 N upward
	22 a)	at A: $8.30\ m\ s^{-1}$; at B: $12.1\ m\ s^{-1}$
	b)	at A: 61.6 N; at B: 208 N
Set 4	6	6×10^{24} kg
	7	1.72×10^{-6} N
	8 a)	2.64×10^{6} m
	8 b)	$8.16\ m\ s^{-2}$ toward the Earth
	8 c)	$7.55 \times 10^{3}\ m\ s^{-1}$
	9	3.80×10^{8} m
	11 b)	2.38×10^{20} N toward the Sun
	18	5.74×10^{3} s (1.59 hours)
	19 a)	$1.37 \times 10^{4}\ m\ s^{-1}$
	19 b)	1.90×10^{27} kg
	20	5.97×10^{24} kg
	21	3.59×10^{7} m
	22	3.05 × (radius of Moon's orbit around Earth)
	23 a)	2.07×10^{22} N; 9.20×10^{21} N
	23 b)	$5.37 \times 10^{4}\ m\ s^{-1}$; $4.39 \times 10^{4}\ m\ s^{-1}$
Set 5	2	120 N m
	3	220 N
	6 b)	18 kg
	8 a)	94 N
	8 b)	447 N
	9	46 kg
	10 a)	630 N
	10 b)	0.75 m toward Q
	11	1.5 m from the front wheels
	12 a)	18 kN
	12 b)	0.375 × length of log from the heavier end